



Title: SOC Alert Management and Incident Response

– Week 2

Name: Dhanavelan

Task: SOC Alert Management and Incident Response

Title

SOC Alert Management and Incident Response

1. Introduction

This document presents the implementation of Security Operations Center (SOC) activities including alert management, incident classification, alert triage, evidence preservation, and incident response. The objective of this task is to simulate real-world SOC operations using structured methodologies and tools. The document demonstrates how alerts are analyzed, prioritized, validated, and handled efficiently to mitigate potential security threats.

2. Theoretical Knowledge

2.1 Alert Priority Levels

Alert priority levels help SOC analysts determine the severity and urgency of security incidents. Critical alerts indicate severe threats such as ransomware attacks or system compromise. High alerts represent serious issues like unauthorized access attempts. Medium alerts include suspicious activities such as brute-force login attempts, while low alerts include minor events like port scanning.

The Common Vulnerability Scoring System (CVSS) is used to measure the severity of vulnerabilities and helps in prioritizing alerts effectively based on risk level and impact.



2.2 Incident Classification

Incident classification involves categorizing security events based on their type, such as malware, phishing, DDoS, insider threats, or data exfiltration. Frameworks like MITRE ATT&CK are used to map incidents to specific techniques and tactics, which helps analysts understand attacker behavior and respond efficiently.

2.3 Incident Response Lifecycle

The incident response lifecycle consists of six phases: Preparation, Identification, Containment, Eradication, Recovery, and Lessons Learned. This structured approach ensures that incidents are handled efficiently, damage is minimized, and future incidents can be prevented through continuous improvement.

3. Alert Management

3.1 Alert Classification Table

The following table represents different types of security alerts categorized based on priority and mapped with MITRE ATT&CK techniques.

<i>Alert ID</i>	<i>Type</i>	<i>Priority</i>	<i>MITRE</i>
1	Phishing	High	T1566
2	Ransomware	Critical	T1486
3	Port Scan	Medium	T1046
4	SSH Brute Force	High	T1110
5	Malware Infection	High	T1204
6	Privilege Escalation	Critical	T1068
7	DDoS Attack	High	T1498
8	Suspicious Login	Medium	T1078



9	Data Exfiltration	Critical	T1041
10	Command Execution	High	T1059

(Caption: Figure 1 – Alert Classification and Prioritization Table)

3.2 Incident Ticket

Title: Critical Alert – Ransomware Detected on Server-X

Date: 20 March 2026

Time: 14:00 IST

Description:

A ransomware attack has been detected on Server-X. A suspicious file named *crypto_locker.exe* was identified. Unauthorized encryption activity was observed, indicating potential system compromise.

Indicators of Compromise (IOCs):

- File: *crypto_locker.exe*
- IP Address: 185.199.108.153
- File Hash:
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b785
2b855

Priority: Critical

Status: Escalated

Assigned To: SOC Analyst

Action Taken:

- System isolated from network
- Malicious process identified and contained
- Initial forensic analysis initiated

3.3 Escalation Email

Subject: Critical Alert Escalation – Ransomware Detected

Dear Tier 2 Team,



A critical ransomware incident has been identified on Server-X. The system shows active encryption behavior linked to `crypto_locker.exe`. The source IP 185.199.108.153 has been flagged as suspicious. Immediate containment actions have been initiated, including isolating the affected system.

Further investigation is required to assess the impact and prevent lateral movement. Kindly prioritize this incident and proceed with advanced analysis.

Regards,

SOC Analyst

4. Alert Triage and IOC Validation

4.1 Alert Triage Table

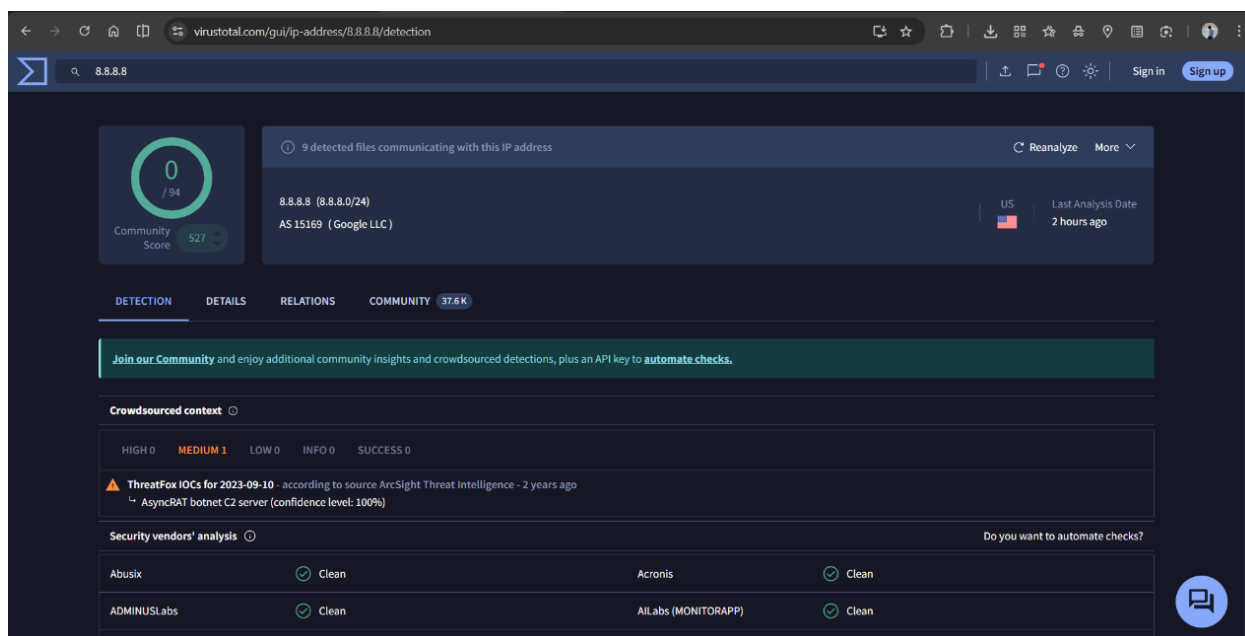
The following table shows alert analysis with priority and investigation status.

Alert ID	Description	IP	Priority	Status
2	SSH Brute Force	185.199.108.153	Medium	Open
3	Port Scan Detected	8.8.8.8	Low	Open
4	Suspicious Login	1.1.1.1	High	Investigating
5	Malware Traffic	8.8.8.8	High	Open
6	Failed Login Attempts	185.199.108.153	Medium	Open

(Caption: Figure 2 – Alert Triage and Status Tracking)

4.2 IOC Validation

The IP address was analyzed using VirusTotal to determine its reputation. The analysis showed no major detections from security vendors, indicating low risk. However, continuous monitoring is recommended to detect any future suspicious activity.



(Caption: Figure 3 – IOC Validation using VirusTotal)

Summary:

The IP address was analyzed using VirusTotal. The results indicate that most security vendors marked the IP as clean, with minimal suspicious activity reported. This suggests a low risk; however, continuous monitoring is recommended to ensure network security and prevent potential threats.

5. Evidence Preservation

The following table documents the collected digital evidence and associated hash values to ensure data integrity.

Item	Description	Date	Hash
Evidence File	evidence.txt	20 March 2026	fb1e17c2023b5d596bbecebbd7ebc6186e052a2bc230a7683f4b66f060cb4839
Memory Dump	Server-X RAM capture	20 March 2026	2aa30034a40effa59c1c931eee558f5d614743df9e4a452fb41667bbd8ad4531
Disk Image	Server-X Disk Backup	20 March 2026	3bbf36f3fe7c97dce87c024583df2c1266de9f333677c481ea14ec747d89cf2b
Log File	Security Logs	20 March 2026	c6cc20482d7ccfa0c3d34296a4d7931bea850d9bc96020044868b65933cb73e8
Network Capture	Packet Capture (PCAP)	20 March 2026	7dabacbf9f1a9807146e703467dc65276d22ed3e32e7a3e661840de321147131

Hash values ensure integrity of evidence and confirm that data has not been altered during investigation.



(Caption: Figure 4 – Evidence Collection and Hash Values)

6. Capstone Project – Incident Response

6.1 Detection Log

Time	Source IP	Alert	MITRE Technique
11:00	185.199.108.153	Web Exploit	T1190
11:05	185.199.108.153	Brute Force	T1110
11:10	185.199.108.153	Suspicious Activity	T1046

6.2 Timeline

11:00 – Web exploit attempt detected
11:05 – Brute force login attempts observed
11:10 – Suspicious scanning activity identified
11:15 – System isolated from network
11:30 – Investigation initiated

6.3 Improved Incident Report

A simulated cyber attack was conducted to evaluate the effectiveness of SOC monitoring and response mechanisms. The attack originated from IP address 185.199.108.153 and involved a web exploitation attempt followed by brute-force login attempts.

The attack was detected through monitoring systems and mapped to MITRE ATT&CK techniques. Immediate response actions were taken, including isolating the affected system and blocking the malicious IP address. Indicators of compromise were analyzed, and evidence was collected for further investigation.



The incident demonstrates the importance of real-time monitoring, rapid detection, and effective response strategies. Continuous improvement in detection systems and proactive security measures are recommended to prevent similar attacks in the future.

6.4 Manager Brief

A security incident involving a potential cyber attack was detected and handled successfully. The affected system was quickly isolated, and the source of the attack was blocked to prevent further impact. No major damage was observed, and the situation was brought under control efficiently.

This incident demonstrates the effectiveness of current monitoring and response processes. However, improvements in early detection and preventive measures are recommended to strengthen overall security. Continuous monitoring and staff awareness will help reduce future risks and ensure system safety.

7. Conclusion

This project demonstrates the implementation of SOC operations including alert management, triage, incident response, and evidence handling. The simulation highlights the importance of structured processes and tools in identifying and mitigating threats. Proper documentation and response strategies are essential for maintaining cybersecurity resilience.

8. References

- MITRE ATT&CK Framework
 - VirusTotal
 - NIST SP 800-61 Incident Handling Guide
 - SANS Incident Handler's Handbook
-